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ARTICLE

SELECTION OF PHASE CHANGE MATERIAL FOR DOMESTIC WATER HEATING USING MULTI CRITERIA DECISION APPROACH

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ABSTRACT

The Selection of appropriate phase change material in the field of domestic water heating system is a difficult task due to the availability of the variety of an enormous number of different materials. The researcher's primary objective is a selection of the desired PCM, which will give the best thermal performance of the domestic water heating latent heat storage system. The paper reports the selection of phase change material with three multi-criteria decision-making (MCDM) methods with conflicting and different objectives. An analytic hierarchy process (AHP) and entropy methods are used to determine the importance criteria weights. For selection purpose, the TOPSIS (Technique for order performance by similarity to ideal solution), VIKOR (Vise KriterijumskaOptimizacijaKompromisnoResenje), and EXPROM2 methods were used, which gives ranking to the alternatives using compromised weight obtained by AHP and entropy methods. This MCDM technique provides a list of all desirable options from the best to the worst suitable materials. Depending on the rankings obtained, the best PCM was found to be the Sodium acetate trihydrate (90%) + graphite (10%), Stearic acid and n-Hexacosane at the given criteria due to its higher physical, chemical, and thermal properties than other alternative PCM materials. The results obtained showed that these methods are feasible tool for the Selection of the best-suited phase change material and can be applied to any material selection decision-making problem to find the best materials.

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Phase Change Material; topsis Method; vikor Method; mcdm; exprom2 Method

1. Introduction

The phase change materials are used to store the thermal energy in the form of latent heat. When phase change materials change its phase from one phase to another, it stores energy in latent heat form. The phase transition temperature almost remains constant during this transition. This phase transition may be solid to liquid, liquid to gas, or vice versa. Out of these, the solid to liquid phase change is most suitable for the latent heat thermal energy storage system (Dinker, Agarwal, and Agarwal 2017). When the melting of PCM takes place from solid to liquid, thermal energy is stored, and when solidification of PCM is carried out from fluid to solid, thermal energy is released.

These latent heat storage materials are classified into organic, inorganic, and eutectic materials (Abdulateef et al. 2018). Organic phase change materials are suitable for a smaller temperature difference with greater latent heat storage capacity per unit volume. These are generally used in low-temperature thermal storage applications due to their lower melting point temperature, which has some excellent thermal properties like non-reactive, non-corrosive, chemically stable, and safe materials. Paraffin and non-paraffin are two categories of organic materials (Reddy, Mudgal, and Mallick 2018).

Kinds of paraffin are hydrocarbon compounds, which consist of hydrogen and carbon molecule with chemical formula C_nH_{2n+2} and, which are derived from coal or petroleum. These PCM possesses some attractive properties such as self-nucleation with no phase segregation, the high latent heat of fusion, low vapour pressure, chemical stability, inert to metal containers, is safe and non-reactive. The melting points of these materials were highly dependent upon the number of atoms of carbon. The melting point and latent heat of fusion are proportional to chain length viz. many carbon atoms. The melting points of Paraffin are a function of a number of carbon atoms. As chain length increases, its latent heat of fusion and melting point also increases. Use of paraffin as PCM restricted due to some adverse properties like low thermal conductivity, incompatibility with plastic containers, partially combustible by specialist containment. At the same time, Non-paraffin are good organic latent heat storage materials with a high latent heat of fusion like fatty acids. These materials offer a broad melting point temperature range; therefore, they can be used in high-temperature applications such as solar water heating systems. Fatty acids have some excellent properties like stability, reliability, and no super cooling than paraffin, but these are corrosive.

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On the other hand, inorganic compounds are divided into salt hydrates and metallic. Salt hydrates are inorganic compounds available in a wide range of melting point temperature from 15°C to 117°C. Salt hydrates consist of many desirable properties, including high thermal conductivity, high latent heat of fusion per unit volume, and less expensive than organic materials. But their corrosive nature restricts the use of these latent heat materials in various applications. Salt hydrates have an affinity to produce an unhydrous salt, which corrodes the container materials. ‚However, another inorganic material-metallics has high thermal conductivity as compared to other material. The metallic compound consists of either low melting point temperature metal or metal eutectics. They are used as storage materials due to their low specific heat, high thermal conductivity, less vapour pressure (Ibrahim et al. 2017).

Another phase change material class is eutectics, which are a mixture of two or more components with fixed freezing/melting points. Eutectics have high storage density, melting point, and high thermal storage capacity with no super cooling effect. In eutectics, both the components melt at the same temperature with or without phase separation during the melting process while they freeze congruently or incongruently.

The selection of suitable phase change material most crucial task of the researcher. Mainly choice of PCM depends upon its melting point. While selecting phase change material, it is vital to lie its melting point in the desired application range. The selection of phase change materials based on collector type i.e. Flat plate & Evacuated tube or concentrating collector, temperature range i.e. Low, Medium & high temperature, and according to application type: (i) TES applications and (ii) thermal insulation applications. In case of TES applications; PCM should have high thermal conductivity with satisfactory melting temperature such as air-conditioning application (melting <15°C), Solar heating (melting 15–90°C) whereas in case of thermal insulation applications preferred to use low thermal conductivity materials like absorption refrigeration (melting >90°C) (Abokersh et al., 2018). However, specific desirable properties required for the selection of LHSM are thermal properties, physical properties, chemical properties, and economic properties.

Desirable properties required for the selection of LHSMs are

1.1. Thermal properties

- Melting point temperature in the desired temperature range
- The high latent heat of fusion
- High thermal conductivity
- High specific heat

1.2. Physical properties

- High density
- Small volume variation during the phase change process
- Low vapour pressure

1.3. Chemical properties

- High chemical stability
- Non-corrosiveness to container
- Non-poisonous, non-flammable and non-explosive

1.4. Economical properties

- Ease of availability,
- Inexpensive,
- Commercially viable
- Availability in large scale

The scientist and researcher are unable to found such material that contains all the desirable properties. The many PCM possesses adverse characteristics that drop their thermal performance in LHTESS. The thermal conductivity of a material is one of the essential properties, which has a more significant influence on meting and solidification time and leads to reduce the performance of the storage unit. Therefore, designers or engineers have to choose the best alternative phase change material with the most significant degree of satisfaction and offer the desired performance at the least cost for all the applicable properties. The selection of the best material for the domestic water heating system from two or more alternative materials; the number of materials selection methods have been developed based on a Multiple Criteria Decision-Making (MCDM) problem. Multiple criteria decision-making (MCDM) refers to making decisions in the presence of usually conflicting multiple criteria.

The selection of materials for engineering applications; some of these methods such as Analytical Hierarchy Process (Jagtap and Bewoor 2017; Singh and Kulkarni 2013; Behera et al. 2019; Li et al. 2018; Baffoe 2019), Entropy method (Kumar and Ray 2014; Zhang 2015), Simple Additive Weighted (SAW) method (Ruby and Banu 2019; Kaliszewski and Podkopaev 2016), TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) (Rastogi et al. 2015; Xu et al. 2017; Rathod and Kanzaria 2011), graph theory and matrix representation approach (GTMA) (Geetha and Sekar 2018), Ashby approach (Srikar and Mark Spearing 2003), Vise KriterijumskaOptimizacijaKompromisnoResenje (V-IKOR) method (Liu et al., 1980–2015, Çalışkan 2013; Opricovic and Tzeng 2004), ELECTRE (elimination and choice expressing the reality) (Brans, Vincke, and Mareschal 1986; Hassan, Rosli, and Redzuan

2018), Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) (Chatterjee and Chakraborty 2012) and COPRAS (complex proportional assessment) (Sennaroglu and Celebi 2018) had been reported in the literature.

The relevant available literature is reviewed to select the desired phase change material for the water heating system.

Saaty (2008) presented a logical procedure to generate priorities and make decisions by collecting relevant information in an organised manner with illustration.Rastogi et al. (2015) selected the phase change material for heating, ventilation, and air-conditioning (HVAC) applications using Ashby approach, Shannon's entropy method, and TOPSIS method. The figure of merits was determined by Ashby approach to obtain PCM performance grade. Then, the weight of this graded PCM is calculated by Shannon's entropy method, and ranking is given using the TOPSIS method. Also, they compared results obtained by TOPSIS with simulation by PCM Express.

Rathod et al. (Rathod and Kanzaria 2011) adopted TOPSIS and Fuzzy TOPSIS methods to select phase change material for the solar domestic hot water system. They used the analytic hierarchy process (AHP) method to determine the weights of the criteria. The results obtained showed that the TOPSIS gives the precise performance rating, while fuzzy TOPSIS gives vague and inaccurate performance ratings. Similarly, the use of AHP and TOPSIS for selecting phase change material for solar air-conditioning units is proposed by Xu et al. (2017). Authors used importance weight criteria by AHP to evaluate the ranking using the TOPSIS method. The results obtained from the order are precise. Therefore, they concluded that the MCDM approach is a handy tool for selecting PCM in latent heat thermal energy systems. A similar type of work is carried out by Socaciu et al. (2016). The paper reported the systematic evaluation of the AHP method to select phase change materials to maintain the vehicle occupant's thermal comfort. They conclude that the use of MCDM also increases system efficiency.

Mao et al. (2018) optimised the residential air-conditioning systems using two multi-criteria decision-making methods to achieve a thermal comfort environment at the lowest energy consumption. The results obtained by the RSM method and TOPSIS method were compared; results by RSM method showed that saving of 74% of computation cost than TOPSIS. Balo and Lütfü (2016) used the analytical hierarchy process (AHP) based on electrical, mechanical, economic, customer, and environment-related criteria to select the 200 W solar panel for a photovoltaic cell. On the other hand,Ozdemir and Sahin (2018) used the analytical hierarchy method to identify the location for installing solar PV cells, considering each location's optimum azimuth angle. The evaluation is carried out based on quantitative and qualitative factors, which have an essential role in producing electricity.

Many researchers applied MCDM methods to choose the materials for automotive components. Liu et al. (1980–2015) develop an evaluation model that depends on the 2-tuple linguistic VIKOR method to select the material for an automotive member and flywheel. The compromised weight of subjective and objective weights of selection criteria is considered an uncertain and imperfect information environment. Also, Kumar and Ray (2014) applied an integrated approach using the entropy method to compute criteria weight and the TOPSIS method to rank the alternative material to identify the suitable material for exhaust manifold.

Calıskan (Çalışkan 2013) examined and selected boron-based coatings for cutting tools using three multi-criteria decision-making methods. The author proposed the organised model to choose the boron-based hard layers used to deposit on cutting tools. The results obtained by EXPROM2, TOPSIS, and VIKOR method give TiBN, and TiSiBN coatings are the best coating material for cutting tool. Similarly, Bhosale, Bhowmik, and Ray (2018) applied the TOPSIS method as the multi-criteria decision-making method to select carbon percentage and copper from the offered options for powder metallurgy. They concluded that 0.8% carbon and 2% copper are the best material composition amongst the proposed alternatives. Likewise, Gangil and Pradhan (2018) examined and optimised EDM process parameters of Titanium alloy using VIKOR and combined RSM (Response Surface Methodology) method. They achieve the best possible results with a minimum error near about 3.107%. A similar approach was considered by Kiani, Liang, and Gross (2018) and developed a comprehensive procedure for repairing concrete structures. To get the best feasible results, they used three different methods to calculate weight: the standard deviation method to evaluate the objective weight, modified digital logic approach to finding subjective weight, and correlation weights methods. The authors suggested using the VIKOR method to obtain the best repair material for concrete structures based on a set of predefined criteria.

In comparison, Chatterjee and Chakraborty (2012) proposed a systematic approach for selecting gear material by applying four preference ranking methods. The results obtained showed that EXPROM2, COPRAS-G, ORESTE, and OCRA methods have great potential to solve complex material selection problems. Also, they express the applicability, effectiveness, and accurateness of all these methods.

Singh and Kulkarni (2013) analysed the critical power-plant equipment using the analytical hierarchy process (AHP) method. Jagtap and Bewoor (2017) carry out similar work proposed using the Analytic Hierarchy Process to identify the thermal power plant's critical equipment. The authors also used the

AHP method to prioritise plant assets according to their criticality. Zhang (Zhang 2015) proposed the TOPSIS method using entropy weight to select the supplier for power grid enterprises reliably. They obtained results that are more rational and persuasive.

Li et al. (2011) determined the safety conditions of coal mines using the TOPSIS Method to use the entropy weight method to identify importance weight. They compared the results obtained by these methods with other evaluation methods like fuzzy synthetic evaluation method, BP artificial neural network, etc. and concluded that the entropy and TOPSIS methods are reasonable to the safety evaluation of coal mines. Brans, Vincke, and Mareschal 1986) develop the process to select and rank the alternatives using the PROMETHEE method. They illustrated the numerical applications- Location for a hydro-electric power plant using PROMETHEE and compared the results with the ELECTRE III method. Sennaroglu et al (Sennaroglu and Celebi 2018) reported the use of AHP integrated PROMETHEE and VIKOR methods to identify the location for military airports. The ranking of the alternative by VIKOR compared with other methods like COPRAS, MAIRCA, and MABAC methods give the same ranking. Firojkhan, Kadam, and Dambhare 2020) evaluate urban green transportation planning based on grey evaluation and entropy-AHP methods.

The present paper deals with selection of appropriate phase change material among the available alternatives for water heating applications. A water heating system using the triplex tube heat exchanger with internal and external fins proposed to investigate the heat transfer enhancement. Figure 1 shows the schematic diagram of the experimental setup of water heating system.

This setup includes three main parts: LH storage unit, storage tank with heater, and other accessories like circulation pump, flow metre, valve, etc. The hot water used for the charging and discharging process. The charging process of water heating unit was facilitated by the heat delivered by the heater. As the temperature of the storage tank reached to melting point temperature of PCM, the circulating pump was activated to deliver hot water through flow metre to melt the PCM in LH storage unit. The temperature of THF controlled by thermostat-electrical heating, which maintain the required set temperature. The PCM discharging process started when the entire PCM melted. The direction of cold water is reversed during discharging process. The final usage of this hot water is for domestic purposes.

The multi-criteria decision method for selecting PCM gives an efficient assessment model, which considers subjectivity, objectivity, uncertainty, and ambiguity for a particular alternative chosen for the present study.

Several criteria should be considered in decision-making between a set of available options to determine the best alternative. In this Paper, Compromised weighting methods used which is combination of analytic hierarchy process (AHP) and Entropy methods to decide the weights of respective criteria, i.e., for latent heat thermal energy storage system, considering different material selection criteria like Latent heat of fusion, Thermal conductivity, specific heat, density, and cost, etc. The AHP method considers the pairwise comparison for the determination weights based on user intellect, experience, and surveys from many users. The entropy method uses the objectivity of the criteria values of all alternatives. So, the combination of AHP and entropy method in the compromised method gives an efficient way to determine the weightage of the respective criteria. The MCDM methods, i.e., TOPSIS, VIKOR, and EXPROM2 method are used for selection of PCM based on ranking given by

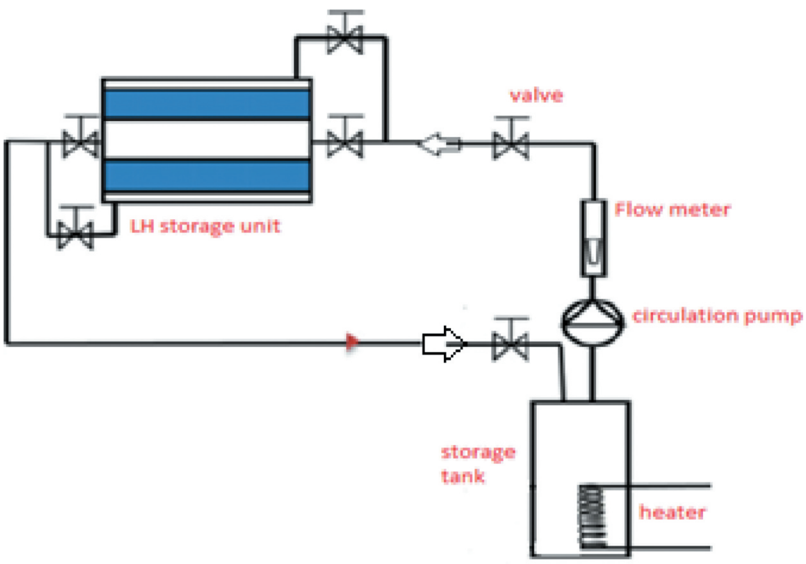


Figure 1. Schematic diagram of the experimental setup of water heating system.

respective method. The Best suitable alternative can be selected for the LHTES in temperature range of 40°C to 70°C.

2. Definition of the decision-making problem

Phase change materials should have some specific properties to maintain their function during charging and discharging. Hence, the selection of the phase change material (PCM) plays a vital role in addition to heat transfer mechanisms. The most of the phase change materials should have a negligible change in its latent heat and melting point. Though the life span of PCM depends on chemical stability, corrosion resistance and thermal stability; while considering alternative phase change materials here more focus given on the melting point temperature range, availability and cost considerations. Considering these three parameters 15 phase change materials were used. The PCMs used here should have a melting point temperature range between 30ºC to 60ºC. The performance rating summary for the 15 alternatives for the eight criteria is summarised in Table 1. The material properties are considered to be latent heat of fusion (LH), thermal conductivity (K) for solid and liquid, specific heat for solid (Cp_s), specific heat for liquid (Cp_l), density (ρ) for solid and liquid state and cost (C) which influences the selection of PCM for a given application. High values of LH, K, Cp, and ρ are desired. The remaining is non-beneficial criteria.The Appendix describes different phase change material used in nano fluid (Table 19) and multiple Phase change material used toghther (Table 20) in Latent Heat storage systems .

In Table 1, the qualitative value of criteria cost is converted into quantitative value using a fuzzy conversion scale based on a ranked value judgement. An eleven-point scale represents the Conversion of linguistic terms into crisp scores using fuzzy logic (Rathod and Kanzaria 2011). The same is presented in Table 2. Whereas Table 3 describes the evaluation matrix, which shows the quantitative value of the criteria with crisp scores.

PCMs are selected for the enhancement of thermal performance latent heat storage systems. Table 1 shows the PCMs used in this study based on the temperature range of 30ºC to 60ºC. Other PCMs are tabulated in Annexure I.

Therefore, the present study employs compromised weighting methods such as AHP and entropy methods to determine the importance weights of evaluation criteria. TOPSIS, VIKOR, and EXPROM2 methods to obtain the performance ranking of the optimal alternatives.

3. Methodology

The methodology implemented in this paper is shown by the flowchart, illustrating how the preference ranking methods can be implemented to select the best

Table 1. Performance Rating Summary.

S.N.	Material	1 LH (KJ/Kg)	2 K(S) (W/mK)	3 K(L) (W/mK)	4 Cp(S) (KJ/KgK)	5 Cp(l) (KJ/KgK)	6 ρ (s) (Kg/m^3)	7 ρ (L) (Kg/m^3)	8 C	Ref.
M1	**P116-Wax**	209	0.14	0.277	2.89	2.89	786	786	High	(Abdulateef et al. 2018,Khan, Khan, and Ghafoor 2016,Salunkhe 2017)
M2	**Stearic acid**	211.6	1.6	0.3	1.76	2.27	940	940	Very high	(Abdulateef et al. 2018)
M3	**Lauric acid**	178	1.6	0.147	1.6	2.27	870	870	Average	(Reddy, Mudgal, and Mallick 2018,Salunkhe 2017)
M4	**Palmitic acid**	201	0.29	0.21	2	2.37	942	862	Average	(Salunkhe 2017)
M5	**n-Eicosane**	248	0.426	0.146	1.926	2.4	910	769	Low	(Reddy, Mudgal, and Mallick 2018,Salunkhe 2017)
M6	**Medicinal paraffin**	146	0.3	2.1	2.25	2.2	830	830	Low	(Salunkhe 2017)
M7	**Paraffin wax**	210	0.24	0.15	2.9	2.1	860	780	Low	(Salunkhe 2017)
M8	**Paraffin wax**	190	0.24	0.22	2	2.15	910	790	Low	(Khan, Khan, and Ghafoor 2016,Salunkhe 2017)
M9	**Stearic acid**	169	0.29	0.29	1.59	1.59	965	847	Average	(Salunkhe 2017)
M10	**Sodium sulphate decahydrate (Glauber's salt)**	180	0.15	0.3	2	2	1460	1458	High	(Salunkhe 2017)
M11	**Sodium acetate trihydrate (90%) + graphite(10%)**	190	2.5	2.5	2.5	2.5	1350	1350	High	(Salunkhe 2017)
M12	**RT55**	172	0.2	0.2	2	2	880	770	Low	(Khan, Khan, and Ghafoor 2016)
M13	**RT60**	167.6	0.2	0.2	2	2	880	770	Very low	(Khan, Khan, and Ghafoor 2016)
M14	**n-Hexacosane**	256	0.21	0.21	2	2	778.3	770	Average	(Reddy, Mudgal, and Mallick 2018)
M15	**RT50**	160	0.2	0.2	2	2	880	760	Low	(Abdulateef et al. 2018,Reddy, Mudgal, and Mallick 2018)

Table 2. Conversion of linguistic terms into fuzzy scores (11-point scale) (Rathod and Kanzaria 2011).

Linguistic term	Crisp Score
Exceptionally low	0.045
Extremely low	0.135
Very low	0.255
Low	0.335
Below average	0.41
Average	0.5
Above average	0.59
High	0.665
Very high	0.745
Extremely high	0.865
Exceptionally high	0.955

phase change material for the domestic water heating system in Figure 2,

4. Determination of the normalised weights

4.1. AHP method

The AHP is one of the most useful and widely used MCDM techniques to solve complex problems involving multiple criteria and formulating and analysing decisions. Saaty 2008) was the first who developed this technique to establish the comparative importance of a set of criteria based on multiple criteria in a hierarchical system. AHP helps to constitute the decision-maker's thoughts and help systematise the problem in a way that is easy to follow and evaluate. AHP allows decision-makers to reproduce a problem in a hierarchical structure in the three steps (Jagtap and Bewoor 2017). These are i) Top-level goal or objective, ii) Criteria used to evaluate the set of alternatives iii) alternatives on the lowest level (Singh and Kulkarni 2013).

AHP has a feature to incorporate tangible as well as non-tangible factors. The goal of the work reported in this paper is to select the phase change material.

The step by step procedure of the AHP method is as follows,

Step 1: Decomposition of the complex decision problem into three levels and construct the hierarchy. The Figure 3 shows the decomposition of the MCDM problem and hierarchy.

Step 2: Determination of the relative judgement of the criteria and alternative within each level. The relative importance of the two criteria is rated using Saaty's 9 point scale. The Saaty's 9 point scale of an index of importance is given in Table 4.

Step 3: Construction of the pair-wise comparison matrix to compare a set of n criteria pair-wise based on their relative importance weights. The pair-wise comparison matrix according to the scale of relative importance represented by an (m x n) evaluation matrix M as,

$$M = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Where, $a_{ij} = 1/a_{ji}\ a_{ji} \neq 0$ and $m = n$. The criteria paired with respect to itself is assigned value $a_{ii} = 1$. The term $a_{ij}(i = 1, 2, 3 \ldots . m, j = 1, 2, 3 \ldots \ldots n)$ is the user-defined rating of comparative importance of criteria i with respect j. The developed individual pairwise comparison matrix is shown below in Table 5,

Step 4: Determination of geometric mean ($\bar{X}_i$) of the ith row and normalised weight (α_i) in the pairwise comparison matrix using the following equations,

$$\bar{X}_i = [a_{i1} \times a_{i2} \times a_{i3} \ldots \ldots \ldots \ldots \ldots . \times a_{in}]^{(1/n)}$$

$$\alpha_i = \frac{\bar{X}_i}{\sum_{j=1}^{j=n} \bar{X}_i}$$

Step 5: Evaluation of the weight sum vector column matrix (A) which is the dot product of the pairwise comparison matrix (M) and normalised weight (α_i). That is $A = M.W$as,

Table 3. Evaluation matrix with crisp score.

	1	2	3	4	5	6	7	8
Material	LH	K(S)	K(L)	Cp(S)	Cp(l)	ρ (s)	ρ (L)	C
M1	209	0.14	0.277	2.89	2.89	786	786	0.665
M2	211.6	1.6	0.3	1.76	2.27	940	940	0.745
M3	178	1.6	0.147	1.6	2.27	870	870	0.5
M4	201	0.29	0.21	2	2.37	942	862	0.5
M5	248	0.426	0.146	1.926	2.4	910	769	0.335
M6	146	0.3	2.1	2.25	2.2	830	830	0.335
M7	210	0.24	0.15	2.9	2.1	860	780	0.335
M8	190	0.24	0.22	2	2.15	910	790	0.335
M9	169	0.29	0.29	1.59	1.59	965	847	0.5
M10	180	0.15	0.3	2	2	1460	1458	0.665
M11	190	2.5	2.5	2.5	2.5	1350	1350	0.665
M12	172	0.2	0.2	2	2	880	770	0.335
M13	167.6	0.2	0.2	2	2	880	770	0.255
M14	256	0.21	0.21	2	2	778.3	770	0.5
M15	160	0.2	0.2	2	2	880	760	0.335

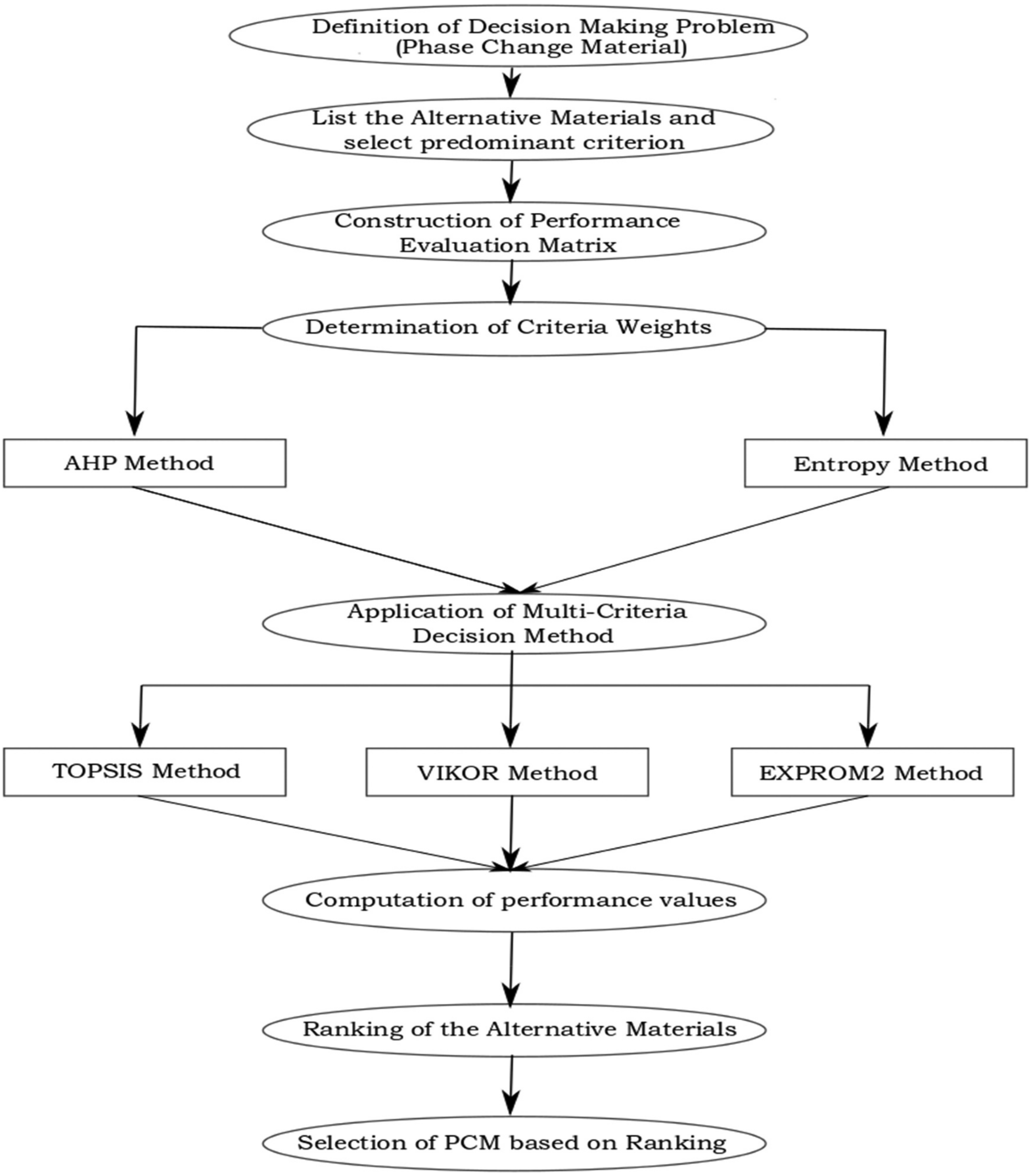


Figure 2. Flowchart for the phase change material selection problem.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_m \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ .. \\ A_m \end{bmatrix}$$

Step 6: Find out the consistency values (CV) of the i^{th} row and average of the consistency values which is nothing but the maximum eigenvalue (λ_{max}) of evaluation matrix M. The consistency values (CV) are calculated as,

$$CV_i = \frac{A_i}{W_i}$$

Step 7: Calculation of the consistency index (CI) to make sure the consistency of the individual view and the precision of the normalised weights. The consistency index is calculated as,

$$CI = \frac{(\lambda_{\max} - m)}{(m - 1)}$$

Step 8: Determination of the consistency Ratio (CR) by,

$$CR = \frac{CI}{RI}$$

where RI is random index. The value of which is determined for different size matrixes, as given in Table 6.

The acceptable value of consistency ratio (CR) is below 0.1 for a reliable result. If this value exceeds 0.1 then to improve consistency, evaluation procedure of pairwise comparison matrix has to be repeated. This consistency ratio gives a basis to evaluate the consistency of user as well as the consistency of overall

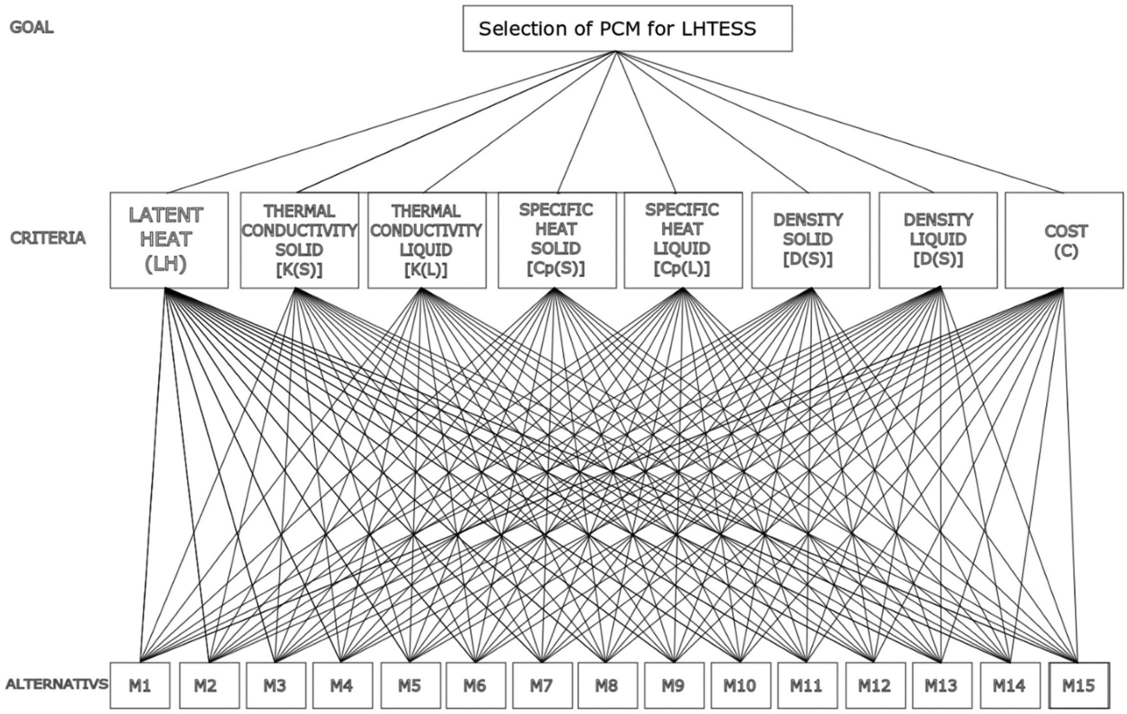


Figure 3. Decomposition of MCDM problem.

Table 4. Relative importance of Criteria (Karayalcin 1980).

Relative importance	Description
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong importance
9	Absolute importance
2,4, 6, 8	Intermediate values

Table 6. Random index (RI) values (Rathod and Kanzaria 2011).

Criteria	RI	Criteria	RI
3	0.52	7	1.35
4	0.89	8	1.4
5	1.11	9	1.45
6	1.25	10	1.49

hierarchy. The consistency ratio obtained by the AHP method is 0.0560 which less than accepted value of 0.1. Hence they are used in the selection process as weights are shown to be consistent. The results are represented in Table 7.

4.2. *Entropy method*

Unlike AHP subjective fixed weight method, entropy is the objective-fixed weight method. The entropy method leads to eliminate human-made disturbances and to make sure that the results are in more agree with facts (Li et al. 2011). This method was first introduced in the theory of information by Shannon. An entropy can measure the degree of disorder, amount of useful information, and its usefulness with the data provided. When the value of entropy is smaller, and if the difference of the alternative values is high, then the relative weight would be higher and vice versa. Therefore, for the determination of weight, this is an objective-fixed weight method. The determination of weight by calculating entropy is to choose the best alternative phase change material for the latent heat thermal energy storage system (Zou, Yun, and Sun 2006). Entropy also deals with the uncertainty in the information devised using the theory of probability (Çalışkan 2013).

The step-by-step procedure of the entropy method is as follows:

Step 1: Construct the performance evaluation matrix (M) of information with m alternative materials and n

Table 5. Pairwise Comparison Matrix.

	LH	K(S)	K(L)	Cp(S)	Cp(l)	D(s)	D(L)	C
LH	1	5	5	7	7	5	5	9
K(S)	0.2	1	1	5	5	0.5	0.5	7
K(L)	0.2	1	1	5	5	0.5	0.5	7
Cp(S)	0.142857	0.2	0.2	1	1	0.2	0.2	3
Cp(l)	0.142857	0.2	0.2	1	1	0.2	0.2	3
D(s)	0.2	2	2	5	5	1	1	7
D(L)	0.2	2	2	5	5	1	1	7
C	0.111111	0.14286	0.14286	0.333333	0.333333	0.142857	0.142857	1
SUM	**2.19683**	**11.54286**	**11.54286**	**29.33333**	**29.33333**	**8.542857**	**8.542857**	**44**

Table 7. Results obtained from the AHP method.

GM ($\bar{X}_i$)	Weight (α)	Eight Sum Vector (A)	Consistency Values (CV)	Consistency Index (CI)	Random Index (RI)	Consistency Ratio (CR)
4.787	0.3942	3.6275	9.2012	0.0784	1.4	0.0560
1.311	0.1080	0.9047	8.3763			
1.311	0.1080	0.9047	8.3763			
0.402	0.0331	0.2808	8.4755			
0.402	0.0331	0.2808	8.4755			
1.855	0.1528	1.2735	8.3372			
1.855	0.1528	1.2735	8.3372			
0.218	0.0180	0.1584	8.8117			
		λ_{max}	**8.54887**			

criteria properties, let $X_{ij}(i = 1, 2, 3 \ldots..m,\ j = 1, 2, 3 \ldots\ldots.n)$is the j^{th} criteria value in the i^{th} alternative.

$$M = \begin{bmatrix} X_{11} & X_{12} & \ldots & X_{1n} \\ X_{21} & \ldots & \ldots & X_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ X_{m1} & X_{m2} & \ldots & X_{mn} \end{bmatrix}$$

This performance evaluation matrix is shown in Table 3.

Step 2: Determine the normalised matrix. The aim of the normalisation is to eliminate influence and get the dimensionless values of different criteria to make the evaluation between them. Normalised matrix N_{ij} is given by,

$$N_{ij} = \frac{X_{ij}}{\sqrt{\sum_{i=1}^{i=m} X_{ij}^2}}$$

Normalised matrix N_{ij} is shown in Table 8.

Step 3: Calculate the entropy value E_j of j^{th} criteria. An entropy value E_j is determined by,

$$E_j = -C \sum_{i=1}^{i=m} P_{ij} \ln(P_{ij})$$

where j = 1, 2, 3, ... n and C = constant that assures the value of E_j is between 0 to 1 and it is given by $C = 1/\ln(m)$, m is a number of alternative materials.

Step 4: Determine the degree of divergence (d_j) of the j^{th} criteria by,

$$d_j = |1 - E_j|$$

Step 5: The entropy weight (βj) of jth criteria is determined by,

$$\beta_j = \frac{d_j}{\sum_{j=1}^{j=n} d_j}$$

After determination of the subjective weights by AHP and objective weights by entropy method, the compromised weights of the criteria (Wj) were calculated as,

$$W_j = \frac{(\alpha_j \times \beta_j)}{\sum_{j=1}^{j=n} (\alpha_j \times \beta_j)}$$

Table 9 shows the criteria weights (αj) obtained using by AHP method and objective criteria weights (βj) obtained by entropy method and compromised weights of the criteria (Wj)

5. Selection of alternative phase change material

5.1. The TOPSIS method

The Technique for Order Performance by Similarity to Ideal Solution (TOPSIS) focuses on multi-criteria problem used to determine the preference ranking of all alternatives which is closest to hypothetically best (positive ideal) solution and farthest away from the hypothetically worst (negative ideal) solution (Zou,

Table 8. Normalised matrix.

	1	2	3	4	5	6	7	8
Material	N1	N2	N3	N4	N5	N6	N7	N8
M1	0.27697	0.04020	0.08232	0.35054	0.33898	0.20973	0.22199	0.34951
M2	0.28041	0.45938	0.08915	0.21348	0.26626	0.25083	0.26549	0.39156
M3	0.23588	0.45938	0.04369	0.19407	0.26626	0.23215	0.24572	0.26279
M4	0.26636	0.08326	0.06241	0.24259	0.27799	0.25136	0.24346	0.26279
M5	0.32865	0.12231	0.04339	0.23361	0.28151	0.24282	0.21719	0.17607
M6	0.19348	0.08613	0.62408	0.27291	0.25805	0.22147	0.23442	0.17607
M7	0.27829	0.06891	0.04458	0.35175	0.24632	0.22948	0.22030	0.17607
M8	0.25179	0.06891	0.06538	0.24259	0.25218	0.24282	0.22312	0.17607
M9	0.22396	0.08326	0.08618	0.19286	0.18650	0.25750	0.23922	0.26279
M10	0.23853	0.04307	0.08915	0.24259	0.23459	0.38958	0.41179	0.34951
M11	0.25179	0.71778	0.74295	0.30324	0.29324	0.36023	0.38129	0.34951
M12	0.22793	0.05742	0.05944	0.24259	0.23459	0.23482	0.21748	0.17607
M13	0.22210	0.05742	0.05944	0.24259	0.23459	0.23482	0.21748	0.13402
M14	0.33925	0.06029	0.06241	0.24259	0.23459	0.20768	0.21748	0.26279
M15	0.21203	0.05742	0.05944	0.24259	0.23459	0.23482	0.21465	0.17607

Yun, and Sun 2006). Hwang and Yoon have first developed this method to convert the multi-criteria system into a referential index. The multi-criteria optimisation problem usually has m alternative materials namely Xi (i =1,2,3, … … m) and n criteria properties, namely Xj (j=1,2,3, … … n).

The step by step procedure to find the TOPSIS index is as follows,

Step 1: Creation of the performance evaluation matrix (M) of the problem with n criteria properties and m alternative materials,

$$M = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1n} \\ X_{21} & \dots & \dots & X_{2n} \\ \dots & \dots & \dots & \dots \\ X_{m1} & X_{m2} & \dots & X_{mn} \end{bmatrix}$$

This performance evaluation matrix is shown in Table 3.

Step 2: Evaluate the normalised matrix based on the root-mean-square value. Normalised matrix N_{ij} is given by,

$$N_{ij} = \frac{X_{ij}}{\sqrt{\sum_{i=1}^{i=m} X_{ij}^2}}$$

where alternatives materials $(i = 1, 2, 3 \ldots..m)$ and criteria properties $(j = 1, 2, 3 \ldots\ldots n)$Normalised weight matrix N_{ij} is shown in Table 8.

Step 3: Build the weighted standardised matrix (V_{ij}). The weighted standardised matrix is obtained by taking the product of a normalised matrix and its associated compromised weights of the criteria (W_j) from Table 9. The weighted standardised matrix is given by,

$$V_{ij} = N_{ij} W_j$$

where $(i = 1, 2, 3 \ldots..m, \quad j = 1, 2, 3 \ldots\ldots n)$ the values of a weighted standardised matrix (V_{ij}) is given in Table 10.

Step 4: Determine the hypothetically best (positive ideal) solution (A_j^*) and hypothetically worst (negative ideal) solution(A_j^-). The best ideal solution and the worst ideal solution for each criterion are depended on their condition of maximisation and minimisation. The hypothetically best (positive ideal) solution (A_j^*) and hypothetically worst (negative ideal) solution (A_j^-) can be expressed as

$$A_1^*, A_2^*, A_3^* \ldots\ldots\ldots, A_n^* = \left\{ (\underbrace{Max}_{i}\ V_{ij}/j \in K), \right.$$

$$\left. (\underbrace{Min}_{i}\ V_{ij}/j \in K')|i = 1, 2, 3 \ldots\ldots.m| \right\}$$

$$A_1^-, A_2^-, A_3^- \ldots\ldots\ldots, A_n^- = \left\{ (\underbrace{Min}_{i}\ V_{ij}/j \in K), \right.$$

$$\left. (\underbrace{Max}_{i}\ V_{ij}/j \in K')|i = 1, 2, 3 \ldots\ldots.m| \right\}$$

where K is the index set of beneficial criteria (larger the-better type) and K' is the index set of non-beneficial criteria-cost criteria (small-the-better type). Table 11 shows the results of best alternative (A*) and worst alternatives (A^-).

Step 5: Determine the distance between the target and best alternative (A*) and distance between the target and worst alternative (A^-). $(S_i{}^*)$represents the

Table 9. Criteria weights by the AHP (*aj*), Entropy (*βj*) and compromised weighting (*Wj*).

Weights	LH	K(S)	K(L)	Cp(S)	Cp(l)	D(s)	D(L)	C
AHP (a_j)	0.3942	0.1080	0.1080	0.0331	0.0331	0.1528	0.1528	0.0180
ENTROPY(β_j)	0.1352	0.1000	0.0944	0.1348	0.1354	0.1346	0.1340	0.1316
WEIGHT(W_j)	**0.4208**	**0.0853**	**0.0805**	**0.0353**	**0.0354**	**0.1624**	**0.1616**	**0.0187**

Table 10. Values of a weighted standardised matrix (V_{ij}).

	1	2	3	4	5	6	7	8
Materials	V1	V2	V3	V4	V5	V6	V7	V8
M1	0.116545	0.003428	0.006627	0.012362	0.012013	0.03405683	0.03588446	0.006531
M2	0.117994	0.039179	0.007177	0.007528	0.009436	0.04072954	0.04291526	0.007316
M3	0.099258	0.039179	0.003517	0.006844	0.009436	0.03769649	0.03971945	0.00491
M4	0.112084	0.007101	0.005024	0.008555	0.009851	0.0408162	0.03935421	0.00491
M5	0.138292	0.010431	0.003493	0.008239	0.009976	0.03942966	0.03510834	0.00329
M6	0.081414	0.007346	0.050242	0.009624	0.009145	0.03596332	0.03789326	0.00329
M7	0.117102	0.005877	0.003589	0.012405	0.008729	0.03726319	0.03561054	0.00329
M8	0.10595	0.005877	0.005263	0.008555	0.008937	0.03942966	0.03606708	0.00329
M9	0.094239	0.007101	0.006938	0.006801	0.006609	0.04181277	0.03866939	0.00491
M10	0.100373	0.003673	0.007177	0.008555	0.008313	0.06326077	0.06656431	0.006531
M11	0.10595	0.061217	0.059812	0.010694	0.010392	0.05849455	0.06163362	0.006531
M12	0.095912	0.004897	0.004785	0.008555	0.008313	0.038,12978	0.03515399	0.00329
M13	0.093459	0.004897	0.004785	0.008555	0.008313	0.03812978	0.03515399	0.002504
M14	0.142753	0.005142	0.005024	0.008555	0.008313	0.03372319	0.03515399	0.00491
M15	0.089221	0.004897	0.004785	0.008555	0.008313	0.03812978	0.03469745	0.00329

Table 11. Best alternative (A*) and worst alternatives (A⁻).

Solution	V1	V2	V3	V4	V5	V6	V7	V8
Best Solution (A*)	0.14275	0.06122	0.05981	0.01240	0.01201	0.06326	0.06656	0.00250
Worst Solution (A⁻)	0.08141	0.00343	0.00349	0.00680	0.00661	0.03372	0.03470	0.00732

Table 12. TOPSIS analysis results.

Materials	S_i^*	S_i^-	RC_i
M1	0.09309	0.03614	0.27967
M2	0.07064	0.05249	0.42630
M3	0.08345	0.04064	0.32748
M4	0.09022	0.03237	0.26404
M5	0.08573	0.05785	0.40289
M6	0.09132	0.04740	0.34168
M7	0.09234	0.03668	0.28427
M8	0.09442	0.02589	0.21519
M9	0.09686	0.01664	0.14660
M10	0.08901	0.04762	0.34854
M11	0.03773	0.09210	0.70941
M12	0.10021	0.01600	0.13765
M13	0.10137	0.01406	0.12180
M14	0.08966	0.06148	0.40677
M15	0.10364	0.01031	0.09051

distance between the target and ideal solution *(A*)*, and (S_i^-) represents the distance between the target and negative ideal solution (A^-).

$$S_i^* = \left\{ \sum_{j=1}^{j=n} (V_{ij} - A_j^*)^2 \right\}^{\frac{1}{2}}$$

$$S_i^- = \left\{ \sum_{j=1}^{j=n} (V_{ij} - A_j^-)^2 \right\}^{\frac{1}{2}}$$

where $i = 1,2,3 \ldots\ldots\, m$ and $j = 1,2,3 \ldots\ldots\, n$

Step 6: Calculate the relative closeness to an ideal solution (RC_i) by,

$$RC_i = \left\{ \frac{S_i^-}{(S_i^* + S_i^-)} \right\} \text{ where } i = 1, 2, \ldots\ldots m \text{ and } 0 \leq RC_i \leq 1$$

The higher values of RC_i mean that better the rank. Therefore, select an alternative material with maximum RC_i. The values obtained for the distance between the target and ideal solution (S_i^*) and target and negative ideal solution (S_i^-) *and* relative closeness to an ideal solution (RC_i) are shown in Table 12.

5.2. The VIKOR method

The VIKOR (Vise KriterijumskaOptimizacijaKompromisnoResenje) is a compromise solution method first introduced by Opricovic and Tzeng (Opricovic and Tzeng 2004). This method concentrates on ranking and selecting the feasible alternative from the set of alternatives in the presence of conflicting criteria. This method gives the compromise solution based on the measure of the closest solution to the ideal solution (Gangil and Pradhan 2018).

The step by step procedure of VIKOR is as follows:

Step 1: Formation of the performance evaluation matrix (M) of data with n criteria properties and m alternative materials,

$$M = \begin{bmatrix} X_{11} & X_{12} & \ldots & X_{1n} \\ X_{21} & \ldots & \ldots & X_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ X_{m1} & X_{m2} & \ldots & X_{mn} \end{bmatrix}$$

This performance evaluation matrix is shown in Table 3.

Step 2: Normalisation of performance evaluation matrix (N_{ij}) based on the root-mean-square value. Normalisation of a matrix is performed by,

$$N_{ij} = \frac{X_{ij}}{\sqrt{\sum_{i=1}^{i=m} X_{ij}^2}}$$

where $(i = 1, 2, 3 \ldots\ldots m)$ are alternative materials and $(j = 1, 2, 3 \ldots\ldots\ldots n)$ are criteria properties. Normalised matrix (N_{ij}) is shown in Table 8.

Step 3: For all criteria; determine the best and the worst values i.e. $(N_{ij})_{\max}$ and $(N_{ij})_{\min}$ based on beneficial criteria and non-beneficial criteria: cost criteria.

Step 4: Find the best (V_i^*) and worst (V_i^-) values of alternative materials $(i = 1, 2, 3 \ldots\ldots m)$ based on maximisation (beneficial criteria (K)) by each criterion $(j = 1, 2, 3 \ldots\ldots\ldots n)$ using,

$$V_1^*, V_2^*, V_3^* \ldots\ldots\ldots\ldots, V_n^* = \left\{ \left(\underbrace{Max}_{i}\; N_{ij}/j \in K\right), \left(\underbrace{Min}_{i}\; N_{ij}/j \in K'\right) | i = 1, 2, 3 \ldots\ldots\ldots m| \right\}$$

And minimisation (non-beneficial criteria cost criteria (K')) by each criterion $(j = 1, 2, 3 \ldots\ldots\ldots n)$ using,

$$V_1^-, V_2^-, V_3^- \ldots\ldots\ldots\ldots, V_n^- = \left\{ \left(\underbrace{Min}_{i}\; N_{ij}/j \in K\right), \left(\underbrace{Max}_{i}\; N_{ij}/j \in K'\right) | i = 1, 2, 3 \ldots\ldots\ldots m| \right\}$$

The best (V_i^*) and worst (V_i^-) values of alternative materials are shown in Table 13

Step 5: Compute the weighted and normalised Manhattan distance (Utility measure) and weighted and normalised Chebyshev distance (Regret measure) by,

Table 13. Best (V_i*) and worst (V_i^-)values of alternative materials.

Solution	N1	N2	N3	N4	N5	N6	N7	N8
Best values (V_i*)	0.33925	0.71778	0.74295	0.35175	0.33898	0.38958	0.41179	0.13402
Worst Values (V_i^-)	0.19348	0.04020	0.04339	0.19286	0.18650	0.20768	0.21465	0.39156

$$S_i = \sum_{j=1}^{j=n} [W_j(V_j{}^{*} - N_{ij})]/(V_j{}^{*} - V_j^-)$$

And,

$$R_i = \underbrace{Max}_{i}\{[W_j(V_j^* - N_{ij})]/(V_j{}^{*} - V_j^-)\}$$

Step 6: Determine the maximum ($S_i{}^*$) and minimum ($S_i{}^-$)values of Utility measure by the following equation,

$$S_i^* = \underbrace{Max}_{i}(S_i) = \underbrace{Max}_{i}\left\{\sum_{j=1}^{j=n}[W_j(V_j{}^{*} - N_{ij})]/(V_j{}^{*} - V_j^-)\right\}$$

$$S_i^- = \underbrace{Min}_{i}(S_i) = \underbrace{Min}_{i}\left\{\sum_{j=1}^{j=n}[W_j(V_j{}^{*} - N_{ij})]/(V_j{}^{*} - V_j^-)\right\}$$

Also, find maximum ($R_i{}^*$) and minimum ($R_i{}^-$) values of Regret measure using,

$$R_i^* = \underbrace{Max}_{i}(R_i) = \underbrace{Max}_{i}\left\{\underbrace{Max}_{i}\{[W_j(V_j^* - N_{ij})]/(V_j{}^{*} - V_j^-)\}\right\}$$

$$R_i^- = \underbrace{Min}_{i}(R_i) = \underbrace{Min}_{i}\left\{\underbrace{Max}_{i}\{[W_j(V_j^* - N_{ij})]/(V_j{}^{*} - V_j^-)\}\right\}$$

Step 7: The values of Q_i are computed by,

$$Q_i = \left\{\frac{\upsilon(S_i - S_i^-)}{(S_i^* - S_i^-)} + \frac{(1-\upsilon)(R_i - R_i^-)}{(R_i^* - R_i^-)}\right\}$$

where υ is nothing but weight for decision-making strategy of the majority of criteria and (1-υ) is the weight for decision-making strategy of individual regret. The value of υ ranges from 0 to 1 (Sennaroglu and Celebi 2018). The three values of υ are generally used; υ = 0.1 represents 'veto', υ = 0.5 corresponds to accepted solution 'consensus', and υ = 0.9 shows consideration of all factors 'majority'. Most of the time value of v assumed to be 0.5. The results obtained using the VIKOR method as tabulated in Table 14.

Step 8: Rank the alternatives according to the obtained values of S_i, R_i and Q_i in ascending order in order to get three ranking lists.

Step 9: Obtained the compromise ranking using the ascending order of Q_i measure of the list for a given v.

Table 14. VIKOR analysis results.

Materials	S_i	R_i	Q_i
M1	0.66060	0.17979	0.33304
M2	0.56903	0.16985	0.23293
M3	0.74933	0.29838	0.63856
M4	0.67774	0.21040	0.40679
M5	0.53184	0.15956	0.18033
M6	0.86142	0.42079	0.97210
M7	0.67511	0.17597	0.33857
M8	0.75787	0.25247	0.55826
M9	0.82771	0.33281	0.77385
M10	0.50243	0.29073	0.40535
M11	0.32814	0.25247	0.17784
M12	0.84472	0.32133	0.76694
M13	0.86460	0.33816	0.81676
M14	0.54062	0.16238	0.19350
M15	0.89294	0.36724	0.89749
Max(S_i*) & (R_i*)	0.89294	0.42079	
Min(S_i^-) & (R_i^-)	0.32814	0.15956	

Step 10: For the given criteria weights, propose a compromise solution (m') which is ranked best with the minimum value of Q_i in the Q_i measure ranking if it satisfies following two conditions.

First condition: 'Acceptable advantage' if $Q(m'')$−$Q(m')$ □1/(m-1) where m'' is the second best ranked alternative by Q_i and m is a number of alternative materials.

Second condition: Adequate stability in decision-making: The alternativz a' must also be the best ranked by S_i and/or R_i.

Step 11. The set of compromised solution is proposed if one of the above conditions given in step 10 is not satisfied. Then, select the alternatives which consist of;

- Alternatives m', m'', …, m^n if the only first condition is not satisfied; m^n is determined by the relation $Q(m^n)$−$Q(m')$ <1/(m-1) which are 'in closeness'.
- Alternative m' and m'' if the only the second condition is not satisfied.

5.3. The EXPROM2 method

The Preference Ranking Organization Methods for Enrichment Evaluations (PROMETHEE) is the outranking method first used by Brans for solving Multi-Attribute Decision-Making problems. This method is used to construct different relation-degrees between alternatives by pair wise comparison. PROMETHEE-I used these different relation-degrees to locate the partial pre-order while PROMETHEE-II used these to set up complete pre-order (Brans, Vincke, and Mareschal 1986). The modified and extended version of the PROMETHEE-II method is an EXPROM2 method that corresponds to the idea of ideal and anti-ideal solutions. It is an interactive multi-criteria decision-making method used to analyse qualitative and quantitative criteria with distinct alternatives. The pair-wise comparisons of alternatives are constructed based on two preference indices. The strict preference (SP) index based on ideal and anti-ideal values and the weak preference (WP) index is based on criteria weights. The perfect measure of the intensity of preference considering all the criterion is achieved by adding a weak preference (WP) index and strict preference (SP) index.

The step by step procedure of **EXPROM2 method** is as (Chatterjee and Chakraborty 2012).

Step 1: Form the performance evaluation matrix (M) of data with n criteria properties and m alternative materials,

$$M = \begin{bmatrix} X_{11} & X_{12} & \ldots & X_{1n} \\ X_{21} & \ldots & \ldots & X_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ X_{m1} & X_{m2} & \ldots & X_{mn} \end{bmatrix}$$

where X_{ij} *(i = 1, 2, ...,m and j = 1, 2, ... n).* Table 3 shows the formed performance evaluation matrix.

Step 2: Normalise the performance evaluation matrix *(M)* by the following equations; for beneficial criteria and non-beneficial criteria.

For beneficial criteria, $r_{ij} = \frac{\{X_{ij} - Min(X_{ij})\}}{\{Max(X_{ij}) - Min(X_{ij})\}}$

And for non-beneficial criteria,

$$r_{ij} = \frac{\{Max(X_{ij}) - X_{ij}\}}{\{Max(X_{ij}) - Min(X_{ij})\}}$$

Step 3: Evaluate the differences in criteria values (d_j) among alternatives pair-wise. The differences in criteria values (d_j) are the difference of i^{th} alternative with respect to other alternatives *(i = 1, 2, ...,m).*

Step 4: Determine the preference function $P_j\ (i,i')$. Preference function indicates that; for *the* j^{th} criterion, the measures of the degree by which i^{th} alternative dominates $(i')^{th}$ alternative. The preference functions are mainly generalised into six types which are tabulated in Table 15. Usual criterion, which is one of the simplest preference functions, is applied here.

Step 5: Determine the weak preference index (WP_j) related to the aggregated preference function considering criteria weights (W_j).

$$WP_j(i,i') = \frac{\left\{\sum_{j=1}^{j=n} W_j \times P_j(i,i')\right\}}{\sum_{j=1}^{j=n} W_j}$$

where (W_j) is the compromised weight of *the* j^{th} criterion.

Step 6: Find the strict preference function $S_j(i,i')$- which depends on the comparison of the maximum difference value (dm_j).

$$S_j(i,i') = \frac{\left[\max(0,(d_j - L_j)\right]}{(dm_j - L_j)}$$

where Limit of preference (L_j) = 0 for usual criterion preference function and (dm_j) = *difference between the ideal and anti-ideal solution of* j^{th} *criterion.*

Step 7: Determine the strict preference index $SP_j(i,i')$. The strict preference index can be evaluated by,

$$SP_j(i,i') = \frac{\left\{\sum_{j=1}^{j=n} W_j \times S_j(i,i')\right\}}{\sum_{j=1}^{j=n} W_j}$$

Step 8: The total preference index $TP_j(i,i')$ can be determined by the following equation as,

$$TP_j(i,i') = Min\left[1,\left\{WP_j(i,i') + SP_j(i,i')\right\}\right]$$

The values of the weak preference index, strict preference index, and total preference index $TP_j(i,i')$ are shown in Table 16.

Step 9: Evaluate the values of Leaving (positive) flow $(\phi^*(i))$ for i^{th} alternative which indicates that how much this alternative dominates the other alternatives. The Leaving (positive) flow is analysed by the following equation,

$$\phi^*(i) = \frac{1}{(m-1)} \sum_{i'=1}^{i'=m} TP_j(i,i') \quad (i \neq i')$$

Step 10: Evaluate the values of Entering (negative) flow $(\phi^-(i))$ for i^{th} alternative which shows that how much this alternative dominated by the other alternatives. The Entering (negative) flow is determined by the following equation,

$$\phi^-(i) = \frac{1}{(m-1)} \sum_{i'=1}^{i'=m} TP_j(i',i) \quad (i \neq i')$$

Where, m is number of alternative materials.

Step 11: Compute the value net outranking flow $(\phi(i))$ which is used to give the complete pre-order to the alternatives. The net outranking flow is given by,

$$\phi(i) = \phi^*(i) - \phi^-(i)$$

Step 12: Depends on the values of net outranking flow $(\phi(i))$ obtained, determine the ranking of all the considered alternatives. Select the best alternative phase change material which has the highest value of net outranking flow $(\phi(i))$.

Table 15. Generalised preference functions(Brans, Vincke, and Mareschal 1986).

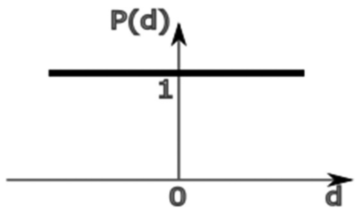
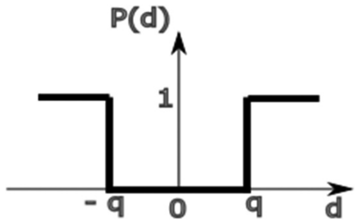
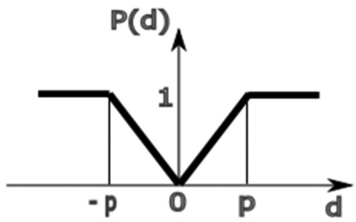
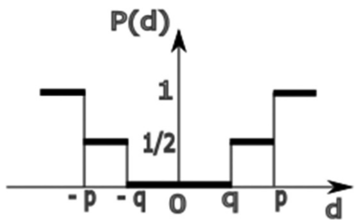
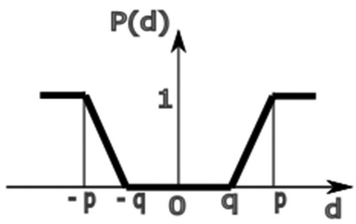
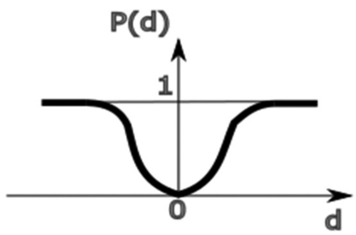
Type of Function	Function Curve	Function Definition
USUAL CRITERIA		$P(d) = \begin{cases} 0 \ id \ d \ = \ 0 \\ 1 \ if \ d \neq 0 \end{cases}$
U-SHAPE CRITERIA		$P(d) = \begin{cases} 0 \, if \ - q \leq d \leq q \\ 1 \ if \ d \langle -q \ or \ d \rangle \ q \end{cases}$
V-SHAPE CRITERIA		$P(d) = \begin{cases} d/p \, if \ - p \leq d \leq p \\ 1 \ if \ d \langle -p \, or \, d \rangle \, p \end{cases}$
LEVEL CRITERIA		$P(d) = \begin{cases} 0 & if \vert d \vert \leq q \\ 1/2 & if \, q \leq d \leq p \\ 1 & if \, p < \vert d \vert \end{cases}$
V-SHAPE WITH PREFERENCE AND INDIFFERENCE CRITERIA		$P(d) = \begin{cases} 0 & if \vert d \vert \leq q \\ \frac{(\vert d \vert - q)}{(p-q)} & if \, q \leq d \leq p \\ 1 & if \, p < \vert d \vert \end{cases}$
GUASSIAN CRITERIA		$P(d) = 1 - e^{-d^2}/2\sigma^2$

Table 16. Weak preference index, strict preference index and total preference index.

Material Pair	$WP_j(i,i')$	$SP_j(i,i')$	$TP_j(i,i')$	Material Pair	$WP_j(i,i')$	$SP_j(i,i')$	$TP_j(i,i')$	Material Pair	$WP_j(i,i')$	$SP_j(i,i')$	$TP_j(i,i')$
1,1	0.00000	0	0	**6,1**	0.48982	0.088798	0.088798	**11,1**	0.48982	0.426271	0.426271
1,2	0.07070	0.047321	0.047321	**6,2**	0.11577	0.074750	0.07475	**11,2**	0.56052	0.326566	0.326566
1,3	0.59069	0.180953	0.180953	**6,3**	0.11577	0.084289	0.084289	**11,3**	1.00000	0.421187	0.421187
1,4	0.59069	0.077321	0.077321	**6,4**	0.20106	0.071728	0.071728	**11,4**	0.57921	0.391678	0.391678
1,5	0.33154	0.06031	0.06031	**6,5**	0.27742	0.089674	0.089674	**11,5**	0.57921	0.425578	0.425578
1,6	0.51018	0.289621	0.289621	**6,6**	0.00000	0.000000	0	**11,6**	1.00000	0.533282	0.533282
1,7	0.29627	0.039852	0.039852	**6,7**	0.36288	0.083162	0.083162	**11,7**	0.54394	0.434251	0.434251
1,8	0.59069	0.131347	0.131347	**6,8**	0.39814	0.083819	0.083819	**11,8**	0.57921	0.429729	0.429729
1,9	0.51018	0.229741	0.229741	**6,9**	0.23650	0.096658	0.096658	**11,9**	1.00000	0.499570	0.499570
1,10	0.49149	0.159156	0.159156	**6,10**	0.23650	0.079162	0.079162	**11,10**	0.65729	0.225508	0.225508
1,11	0.49149	0.093812	0.093812	**6,11**	0.00000	0.000000	0	**11,11**	0.00000	0.000000	0.000000
1,12	0.75233	0.208682	0.208682	**6,12**	0.39814	0.094670	0.09467	**11,12**	1.00000	0.516582	0.516582
1,13	0.75233	0.228564	0.228564	**6,13**	0.41683	0.097720	0.09772	**11,13**	1.00000	0.536464	0.536464
1,14	0.49392	0.062343	0.062343	**6,14**	0.56052	0.106281	0.106281	**11,14**	0.57921	0.464955	0.464955
1,15	0.75233	0.256902	0.256902	**6,15**	0.39814	0.096985	0.096985	**11,15**	1.00000	0.564802	0.564802
2,1	0.92930	0.138893	0.138893	**7,1**	0.70373	0.025335	0.025335	**12,1**	0.24767	0.024559	0.024559
2,2	0.00000	0	0	**7,2**	0.03527	0.030689	0.030689	**12,2**	0.03527	0.006461	0.006461
2,3	0.87927	0.1803	0.1803	**7,3**	0.53656	0.157511	0.157511	**12,3**	0.27815	0.014963	0.014963
2,4	0.76691	0.118375	0.118375	**7,4**	0.45606	0.058657	0.058657	**12,4**	0.00000	0.000000	0.000000
2,5	0.50851	0.110075	0.110075	**7,5**	0.27742	0.028905	0.028905	**12,5**	0.27742	0.004070	0.004070
2,6	0.88423	0.367144	0.367144	**7,6**	0.61844	0.269468	0.269468	**12,6**	0.58317	0.111370	0.111370
2,7	0.96473	0.136778	0.136778	**7,7**	0.00000	0.000000	0	**12,7**	0.24289	0.006474	0.006474
2,8	0.96473	0.195302	0.195302	**7,8**	0.45606	0.100736	0.100736	**12,8**	0.00000	0.000000	0.000000
2,9	0.83762	0.264638	0.264638	**7,9**	0.49149	0.206008	0.206008	**12,9**	0.49149	0.033690	0.033690
2,10	0.56020	0.183694	0.183694	**7,10**	0.57678	0.144968	0.144968	**12,10**	0.08529	0.001807	0.001807
2,11	0.43948	0.085679	0.085679	**7,11**	0.45606	0.087276	0.087276	**12,11**	0.00000	0.000000	0.000000
2,12	0.96473	0.282155	0.282155	**7,12**	0.73843	0.176080	0.17608	**12,12**	0.00000	0.000000	0.000000
2,13	0.96473	0.302038	0.302038	**7,13**	0.75711	0.195962	0.195962	**12,13**	0.43948	0.019882	0.019882
2,14	0.54394	0.1479	0.1479	**7,14**	0.48002	0.049815	0.049815	**12,14**	0.16238	0.024225	0.024225
2,15	0.96473	0.330376	0.330376	**7,15**	0.73843	0.224300	0.2243	**12,15**	0.58244	0.048220	0.048220
3,1	0.40931	0.092225	0.092225	**8,1**	0.40931	0.034077	0.034077	**13,1**	0.24767	0.024559	0.024559
3,2	0.00000	0	0	**8,2**	0.03527	0.006461	0.006461	**13,2**	0.03527	0.006461	0.006461
3,3	0.00000	0	0	**8,3**	0.69894	0.068697	0.068697	**13,3**	0.27815	0.014963	0.014963
3,4	0.24693	0.049194	0.049194	**8,4**	0.08051	0.000342	0.000342	**13,4**	0.00000	0.000000	0.000000
3,5	0.34612	0.072143	0.072143	**8,5**	0.27742	0.009386	0.009386	**13,5**	0.27742	0.004070	0.004070
3,6	0.88423	0.196384	0.196384	**8,6**	0.58317	0.187372	0.187372	**13,6**	0.58317	0.094538	0.094538
3,7	0.46344	0.0833	0.0833	**8,7**	0.43997	0.017983	0.017983	**13,7**	0.24289	0.006474	0.006474
3,8	0.30106	0.077239	0.077239	**8,8**	0.00000	0.000000	0	**13,8**	0.00000	0.000000	0.000000
3,9	0.73843	0.105902	0.105902	**8,9**	0.49149	0.106636	0.106636	**13,9**	0.07070	0.022214	0.022214
3,10	0.12073	0.059761	0.059761	**8,10**	0.54152	0.045595	0.045595	**13,10**	0.08529	0.001807	0.001807
3,11	0.00000	0	0	**8,11**	0.00000	0.000000	0	**13,11**	0.00000	0.000000	0.000000
3,12	0.72185	0.110357	0.110357	**8,12**	0.94605	0.086853	0.086853	**13,12**	0.00000	0.000000	0.000000
3,13	0.72185	0.130239	0.130239	**8,13**	0.96473	0.106735	0.106735	**13,13**	0.00000	0.000000	0.000000
3,14	0.44475	0.102594	0.102594	**8,14**	0.52526	0.041518	0.041518	**13,14**	0.16238	0.024225	0.024225
3,15	0.72185	0.158577	0.158577	**8,15**	0.94605	0.135073	0.135073	**13,15**	0.58244	0.031389	0.031389
4,1	0.40931	0.060181	0.060181	**9,1**	0.48982	0.062630	0.06263	**14,1**	0.50608	0.182322	0.182322
4,2	0.23309	0.009663	0.009663	**9,2**	0.16238	0.005955	0.005955	**14,2**	0.45606	0.176307	0.176307
4,3	0.73438	0.120783	0.120783	**9,3**	0.24289	0.027520	0.02752	**14,3**	0.53656	0.311302	0.311302

(*Continued*)

Table 16. (Continued).

Material Pair	$WP_j(i,i')$	$SP_j(i,i')$	$TP_j(i,i')$	Material Pair	$WP_j(i,i')$	$SP_j(i,i')$	$TP_j(i,i')$	Material Pair	$WP_j(i,i')$	$SP_j(i,i')$	$TP_j(i,i')$
4,4	0.00000	0	0	**9,4**	0.24289	0.008215	0.008215	**14,4**	0.42079	0.210396	0.210396
4,5	0.45848	0.039633	0.039633	**9,5**	0.42322	0.042381	0.042381	**14,5**	0.71689	0.041307	0.041307
4,6	0.79894	0.255411	0.255411	**9,6**	0.76350	0.130370	0.13037	**14,6**	0.43948	0.427083	0.427083
4,7	0.54394	0.056034	0.056034	**9,7**	0.50851	0.053414	0.053414	**14,7**	0.51998	0.184311	0.184311
4,8	0.88423	0.080472	0.080472	**9,8**	0.50851	0.036794	0.036794	**14,8**	0.43948	0.258767	0.258767
4,9	0.65314	0.158186	0.158186	**9,9**	0.00000	0.000000	0	**14,9**	0.49149	0.355021	0.355021
4,10	0.54152	0.095478	0.095478	**9,10**	0.08529	0.005059	0.005059	**14,10**	0.50608	0.292897	0.292897
4,11	0.42079	0.042,079	0.042,079	**9,11**	0.00000	0.000000	0	**14,11**	0.42079	0.252475	0.252475
4,12	0.96473	0.166983	0.166983	**9,12**	0.50851	0.050702	0.050702	**14,12**	0.60527	0.328327	0.328327
4,13	0.96473	0.186865	0.186865	**9,13**	0.92930	0.059108	0.059108	**14,13**	0.60527	0.348209	0.348209
4,14	0.44475	0.073277	0.073277	**9,14**	0.48982	0.067931	0.067931	**14,14**	0.00000	0.000000	0.000000
4,15	0.96473	0.215203	0.215203	**9,15**	0.92930	0.087446	0.087446	**14,15**	0.76691	0.376547	0.376547
5,1	0.66846	0.189062	0.189062	**10,1**	0.48982	0.317320	0.31732	**15,1**	0.24767	0.024559	0.024559
5,2	0.49149	0.147256	0.147256	**10,2**	0.35929	0.250286	0.250286	**15,2**	0.03527	0.006461	0.006461
5,3	0.65388	0.289624	0.289624	**10,3**	0.87927	0.306654	0.306654	**15,3**	0.27815	0.014963	0.014963
5,4	0.54152	0.185525	0.185525	**10,4**	0.42322	0.270782	0.270782	**15,4**	0.00000	0.000000	0.000000
5,5	0.00000	0	0	**10,5**	0.45848	0.310415	0.310415	**15,5**	0.11577	0.003839	0.003839
5,6	0.70390	0.41925	0.41925	**10,6**	0.76350	0.438148	0.438148	**15,6**	0.58317	0.065465	0.065465
5,7	0.70390	0.172174	0.172174	**10,7**	0.42322	0.317649	0.317649	**15,7**	0.24289	0.006474	0.006474
5,8	0.54152	0.235408	0.235408	**10,8**	0.42322	0.301029	0.301029	**15,8**	0.00000	0.000000	0.000000
5,9	0.57678	0.338245	0.338245	**10,9**	0.91471	0.330335	0.330335	**15,9**	0.07070	0.022214	0.022214
5,10	0.54152	0.281004	0.281004	**10,10**	0.00000	0.000000	0	**15,10**	0.08529	0.001807	0.001807
5,11	0.42079	0.221872	0.221872	**10,11**	0.32403	0.051213	0.051213	**15,11**	0.00000	0.000000	0.000000
5,12	0.70390	0.316946	0.316946	**10,12**	0.84401	0.344094	0.344094	**15,12**	0.00000	0.000000	0.000000
5,13	0.72258	0.336828	0.336828	**10,13**	0.84401	0.363976	0.363976	**15,13**	0.01869	0.003051	0.003051
5,14	0.28311	0.050081	0.050081	**10,14**	0.42322	0.331082	0.331082	**15,14**	0.16238	0.024225	0.024225
5,15	0.86554	0.364934	0.364934	**10,15**	0.84401	0.392314	0.392314	**15,15**	0.00000	0.000000	0.000000

Table 17. EXPROM2 analysis results.

Materials	$\phi^{+}(i)$	$\phi^{-}(i)$	$\phi(i)$
M1	0.66297	0.59717	0.06580
M2	0.89034	0.26573	0.62461
M3	0.51985	0.51985	0.00000
M4	0.71853	0.71853	0.00000
M5	0.82380	0.46874	0.35506
M6	0.38251	0.87426	−0.49174
M7	0.64840	0.57116	0.07724
M8	0.54292	0.56061	−0.01768
M9	0.49319	0.67835	−0.18516
M10	0.81959	0.47803	0.34156
M11	0.97393	0.27199	0.70194
M12	0.26581	0.76381	−0.49800
M13	0.19972	0.83740	−0.63768
M14	0.78807	0.52014	0.26793
M15	0.14379	0.89877	−0.75498

The results obtained by the EXPROM2 method are tabulated in Table 17.

Table 18. Rankings of the alternative materials.

S. N.	Materials	TOPSIS	VIKOR	EXPROM2
1	P116-Wax	9	5	7
2	Stearic acid	2	4	2
3	Lauric acid	7	10	8
4	Palmitic acid	10	8	8
5	n-Eicosane	4	2	3
6	Medicinal paraffin	6	15	12
7	Paraffin wax	8	6	6
8	Paraffin wax	11	9	10
9	Stearic acid	12	12	11
10	Sodium sulphate decahydrate	5	7	4
11	Sodium acetate trihydrate (90%) + graphite(10%)	1	1	1
12	RT55	13	11	13
13	RT60	14	13	14
14	n-Hexacosane	3	3	5
15	RT50	15	14	15

6. Results and discussion

The present study applied MCDM methods like TOPSIS, VIKOR, and EXPROM2 on the Phase change material selection problem. These methods use AHP and Entropy criterion weighting methods to find the weights of the criterion. The compromised weight determined by AHP and Entropy method is applied to these MCDM methods and used to select the phase change material. The results obtained by the applied methods- TOPSIS, VIKOR, and EXPROM2 are discussed in the following sections,

6.1. TOPSIS method

Firstly, the distance between the target and the best alternative and target and worst alternative were computed. The relative closeness to the ideal solution was measured, and rank is given to alternative materials according to relative closeness measures. The performance ratings are shown in Table 11. The alternative material is to be ranked 1 with the highest value of RC_i. Therefore, select an alternative material with maximum RC_i. It is concluded from the descending order of ranking of alternative materials: the best alternative phase change material is with the maximum value of 0.70941designated as M11. The best alternative material M11 is the best choice for a given purpose when eight criteria were taken into consideration and also has a high value of latent heat, thermal conductivity, specific heat, and density.

6.2. VIKOR method

The best and worst alternatives of all criteria were estimated from the performance evaluation matrix in this method. After getting the best (V_i^*) and worst (V_i^-) alternatives, the values of S_i, R_i and Q_i were determined. Rank the alternatives material according to the obtained values of Si, Ri, and Qi in ascending order to get three ranking lists. In the present study, a compromise alternative is proposed based on best with the minimum value

Figure 4. Comparison of rankings obtained with the MCDM methods.

of Q_i in the Q_i measure ranking doesn't satisfy the given first condition. Therefore a set of compromised solutions is proposed. The chosen alternative materials by the VIKOR ranking method are designated as M11, M5, and M14, tabulated in 2Table 13.

6.3. EXPROM2 method

The EXPROM2 method employs the preference functions for the computation of the total pre-order of the alternatives. The normalisation of the performance evaluation matrix was done first. Using the compromised weight obtained by AHP and Entropy method, the corresponding weak and strong preference functions were calculated. Then, consequent weak and strong preference indices were computed with considering the usual criterion. The positive (leaving) and the negative (entering) outranking flows were estimated after that. Depends on these positive (leaving) values and the negative (entering) outranking flows, related net outranking flows were calculated. The final best rank is assigned to the alternative material, which has the highest value of net outranking flow. The ranking of the results by the EXPROM2 method shows that the alternative material M11 has the highest value of 0.70194 of net outranking flow. Therefore, the material designated by M11 was found the best phase change material due to its high value of thermal conductivity, latent heat of fusion, and density.

6.4. Comparison of the results

The results obtained by the three MCDM methods; TOPSIS, VIKOR, and EXPROM2 were compared to select the best-suited phase change material for given application. The rankings obtained by these three methods are tabulated in Table 18.

The Table 18 represents the comparison of ranking of PCM with respect to MCDM method. For TOPSIS method; the materials Sodium acetate trihydrate (90%) + graphite (10%), Stearic acid, and n-Hexacosane ranked 1, 2 and 3, respectively, whereas materials RT55, RT60, RT50 ranked as 13, 14, 15, respectively. Therefore the materials Sodium acetate trihydrate (90%) + graphite (10%), Stearic acid, and n-Hexacosane are first, second and third best alternatives materials used and RT55, RT60, RT50 are worst alternative materials for given criterion. In case of VIKOR method Sodium acetate trihydrate (90%) + graphite(10%), n-Eicosane and n-Hexacosane ranked best three whereas RT55, RT60, RT50, ranked worst three materials. The ranking obtained by EXPROM2 method showed that Sodium acetate trihydrate (90%) + graphite (10%), Stearic acid, n-Eicosane ranked 1, 2, 3 and RT55, RT60, RT50 ranked 13, 14, 15, respectively.

Figure 4 shows comparison of rankings obtained with three MCDM methods. The ranking results obtained by three methods showed that the alternative materials Sodium acetate trihydrate (90%) + graphite (10%), Stearic acid, and n-Hexacosane are the best phase change materials for the latent heat thermal energy storage system due to their best thermal, physical and chemical properties and materials designated as RT55, RT60, RT50 are not suitable for given criterion.

The Pearson correlation coefficient of for comparing the above three methods shows the between TOPSIS and VIKOR method p value is 0.757, TOPSIS and EXPROM2 Method p value is 0.898, and VIKOR and EXPROM2 0.923 which is acceptable for the selection of material also all three methods gives the best alternative to Sodium acetate trihydrate (90%) + graphite(10%).

7. Conclusions

The selection of the phase change material for the domestic water heating system is one of most important tasks. To use the latent heat thermal energy storage system most efficiently, the selection of proper phase change material plays a very crucial role. Many of the researchers choose phase change material based on their experience, availability and cost. On the other hand, in this paper phase change material is selected based on many different contradictory criteria with a number of alternatives. The present study suggests three MCDM methods-TOPSIS, VIKOR and EXPROM2, in solving the PCM selection problem. All the three methods use compromised weight of AHP and entropy method to assign to the criteria which are used for selection of PCM.

The ranking obtained using these three MCDM methods match with each other. The TOPSIS, VIKOR and EXPROM2 methods lead to choice of M11 i.e. Sodium acetate trihydrate (90%) + graphite (10%) as a preferred alternative at the given criteria. The materials designated by M2, M14 and M5 are the second, third and fourth best alternatives materials used. Other than these four alternatives, the preferences vary between the three methods. The Sodium acetate trihydrate (90%) + graphite (10%) material obtained the best rank due to its higher thermal and physical properties than other alternative PCM materials. The material of container should be non corrosive to Sodium acetate trihydrate (90%) + graphite (10%). The systematic evaluation of the MCDM problem can reduce the risk of a poor selection of the material. In short, the proposed methodology provides feasible tool and systematic approach to narrow down the number of alternatives and can be applied to any material selection decision-making problem to find the best materials.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix I

Table 19. PCM Based on Nano Fluid additives.

S.N.	PCM	Additives
1	Paraffin	Grapheme And Exfoliated Graphite Sheet
2	Erythritol	Carbon Fibres
3	Paraffin And Ethylene Vinyl Acetate	Expanded Grapheme And Carbon Fibre
4	Beeswax	Graphene
5	Palmitic– Stearic Acid Eutectic Mixture	Carbon Nanotubes
6	Paraffin	Graphene Nano-Platelets (GNPs), Carbon Nanofibers (CNFs), And Multi-Wired Carbon Nanotubes
7	Octadecane	CNT And Graphene
8	Palmitic Acid (PA)	GrapheneNanoplatelets
9	Palmitic Acid	Multi Walled Carbon Nanotube
10	Paraffin Wax	Multi-Walled Carbon Nanotubes
11	Wax	Carbon Nanotubes (SWCNTs), Multiwall CNTs, And Carbon Nanofibers
12	N-Octadecane	Titanium Oxide Nanoparticles
13	Eicosane	Silver Nanoparticles
14	Paraffin	Aluminium Foam
15	Pure Water	Al2O3 Nanofluid
16	Eutectic Of Lithium Carbonate And Potassium Carbonate	SiO2 Nanoparticles
17	Paraffin Wax	Nano-Al2O3
18	n-Octadecane	MPSiO2
19	Icosane Wax	Copper Foam And Copper Matrix
20	Paraffin Wax	Aluminium Foams
21	Pure Eicosane	CuO Nanoparticles
22	Erythritol	Spherical Graphite (SG), Expanded Graphite (EG) And Nickel Particles
23	Paraffin	Al2O3
24	Paraffin Wax	Aluminium Powder
25	Water	Copper Nanoparticle
26	Polyethylene	Hybrid Grapheme Aero Gels
27	N-Octadecane And Stearic Acid Eutectics	Hexagonal Boron Nitride
28	Paraffin Wax (RT20 And RT25)	Alumina (Al2O3) And Carbon Black (CB)
29	Paraffin	Hexagonal Boron Nitride
30	Tetradecanol	Copper Nanowires

Table 20. Multiple PCM used Together.

Sr.No.	No. of PCMs	PCMs Used
1	2	P116 and n-octadecane
2	3	$CaCl_2.6H_2O$, Paraffin C18, RT25
3	3	80.5LiF–19.5CaF_2, Haynes 188, 37.3He–62.7Xe
4	2	d-mannitol and hydroquinone.
5	3	KNO_3, KNO_3 + KCl, $NaNO_3$
6	3	NaOH + NaCl, KCl + MnCl2 + NaCl, NaOH + NaCl + Na2CO3
7	5	LiF + CaF_2, LiF + MgF_2,
8	5	LiF/CaF_2, others not available
9	3	paraffin 40, paraffin50 and paraffin 60
10	2	Capric-Lauric acid eutectic and Oleic acid
11	3	RT18,RT20 and RT25
12	3	inorganic HS-W1,organic HS-W2 and Paraffin C15